PROJECT RESOURCING AND LOADING PLAN

MBA 666

Green Hydrogen Production Plant

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1MW Green Hydrogen Plant **Document Purpose:** To identify, allocate, and schedule all key resources required to complete the project with a primary resource control focus on the critical period of **Weeks 1-27**.

1.Key Activity Data & Constraints

The project is **Time-Constrained** with a primary resource control focus on the critical period of **Weeks 1-27.**We utilize resource levelling to reduce peak resource demands and minimise fluctuations in resource use.

From PROJECT TIMELINES AND BASELINE SCHEDULING document we get key activity data:

ID	Duration	ES (Wk)	EF (Wk)	LS (Wk)	LF (Wk)	Slack	Required Human Resource
A	4	0	4	0	4	0	Specialist Team
B	6	4	10	10	16	6	Civil Team (HR-C)
C	12	4	16	4	16	0	Specialist Team
D	8	4	12	14	22	10	Specialist Team
E	14	4	18	12	26	8	Install Team (HR-I)
F	10	16	26	16	26	0	Install Team (HR-I)
G	4	12	16	22	26	10	Specialist Team
H	12	26	38	26	38	0	Specialist Team

I	4	38	42	38	42	0	Specialist Team
J	2	42	44	42	44	0	Specialist Team
K	2	44	46	44	46	0	Specialist Team

2.Analysis of Un-Levelled Schedule

The critical path is A->C->F->H->I->J(Duration: 36-weeks). The maximum demand for our field teams(HR-C and HR-I) is always 1 team concurrently,but the schedule is volatile.We optimise resource levelling for the peak 27 weeks where resourcing conflict can occur.

Resource Load before levelling

Week	Activity	Resource Requirement
1-4	A	0
5-10	B	HR-C(1 team)
11	-	0
12	G	HR-I(1 team)
13-16	G	HR-I(1 team)
17-26	F	HR-I(1 team)
27+	H	0

The **1-week gap (Wk 11)** where resource demand is 0 is inefficient. We must shift Activity B to fill this gap.This gap creates an inefficient use of labor, leading to higher indirect costs (e.g., equipment storage, personnel time) and logistics issues

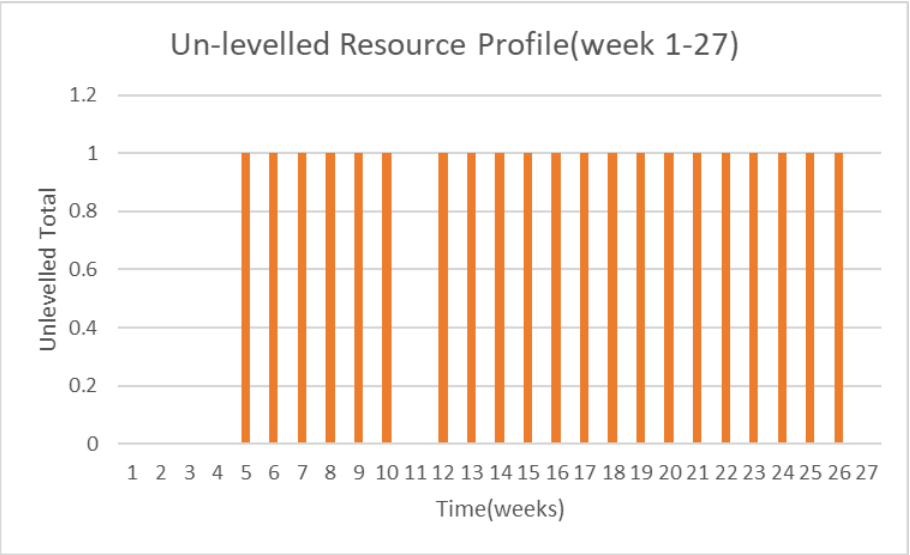

Fig:Unevelled Resource Profile

3. Resource Levelling Calculation

We use the slack of Activity B (SLK = 6 Wk) to delay its start.The levelling process targets Activity B because it has **6 weeks of slack**. This slack is the amount of time B can be delayed without affecting the overall 36-week project deadline., smoothing the handover to the Install Team (HR-I).

Resource	Original ES	Original EF	Slack (SLK)	Required Delay	New ES	New EF	LS
B(HR-C)	4	10	6	1 week	5	11	10
G(HR-I)	12	16	6	0	12	16	22
F(HR-I)	16	26	0	0	16	26	16

- **B** now finishes at **Week 11**.
- **G**'s predecessor **B** is satisfied at **Week 11**. **G**'s other predecessor **D** finishes at **Week 12**.
- The new **ES** for **G** remains **Week 12** (Max of 11 and 12).

By delaying Activity B by 1 weeks (using only a fraction of its SLK), we achieve a continuous workflow with no idle time between HR-C and HR-I work.

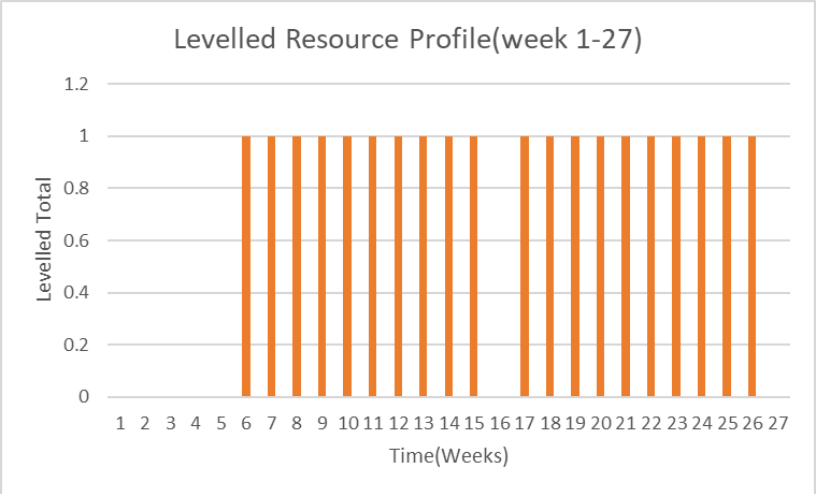

Fig:Levelled Resource Profile

The **Levelled Total** remains 1 or less throughout the entire Wk 1-27 critical period. The original idle time at Wk 11 is successfully eliminated, although it is replaced by a planned delay at Wk 5 and an unavoidable gap at Wk 16$

4. Analysis of Levelled Schedule

The new schedule utilizes resources continuously from **Week 6** to **Week 26**, eliminating the mid-project idle time.

Resource Load after levelling

Week	Activity	Resource Requirement
1-4	A	0
5	-	0
6-11	B	HR-C(1 team)
12	G	HR-I(1 team)
13-15	G	HR-I(1 team)
16	-	0
17-26	F	HR-I(1 team)
27+	H	0

Week 5: Now we have a **1-week early** delay before B starts. This is a controlled, planned idle period that is much more efficient than a reactive mid-project gap

Week	HR-C	HR-I	Unlevelled Total	Levelled HR-C	Levelled HR-I	Levelled Total
1	0	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	1	0	1	0	0	0
6	1	0	1	1	0	1
7	1	0	1	1	0	1
8	1	0	1	1	0	1
9	1	0	1	1	0	1
10	1	0	1	1	0	1
11	0	0	0	1	0	1
12	0	1	1	0	1	1
13	0	1	1	0	1	1
14	0	1	1	0	1	1
15	0	1	1	0	1	1
16	0	1	1	0	0	0
17	0	1	1	0	1	1
18	0	1	1	0	1	1
19	0	1	1	0	1	1
20	0	1	1	0	1	1
21	0	1	1	0	1	1
22	0	1	1	0	1	1
23	0	1	1	0	1	1
24	0	1	1	0	1	1
25	0	1	1	0	1	1
26	0	1	1	0	1	1
27	0	0	0	0	0	0

Fig Resource loading Profile

5. Contingency Plan: Resource Constraint Heuristics

Since the project is currently time-contrained any risk event could force the projects into a resource constrained environment.
If this happens ,project team cannot level and should implement the parallel method using specific heuristic priority rules to manage resource conflicts.This serves as the primary **mitigation strategy** for resource loss.

Priority Rank	**Heuristic Rule**	**Purpose**
1	Minimum Slack	Prioritizes the most time-critical activity to protect the overall project completion time.
2	Smallest Duration	Prioritizes the fastest job to clear the limited resource bottleneck quickly.
3	Lowest Activity ID Number	Used as a final, objective tie-breaker.